

TEAM 1

AI Sign Language Translator

Converts sign language into speech, in real time.



TEAM LEADER

Gouri Dehare



TEAM MEMBERS

1. Om mourya 2. sakshi khante 3. shamli zalke



SELECTED AI AGENT DOMAIN

Accessibility & Assistive Communication
(Computer Vision + NLP + Speech Synthesis)





Slide 2

Problem Statement

Over 70 million deaf and hard-of-hearing people worldwide rely on sign language as their primary means of communication — yet most hearing people cannot understand it, creating a daily communication barrier.



What is the problem?

No real-time, low-cost tool exists that instantly converts sign language into spoken language, so communication between signers and non-signers stays slow and dependent on interpreters.



Who experiences this?

Deaf and hard-of-hearing individuals, their families, teachers, doctors, and employers who don't know sign language.



Where does it occur?

Hospitals, schools, government offices, workplaces, retail counters, and everyday public interactions.



Current pain point

Human interpreters are expensive (₹500–₹2,000/hr) and often unavailable on short notice, leaving urgent needs unmet.

Goal: Deliver a specific, measurable improvement — cut translation wait time from minutes (finding an interpreter) to under 2 seconds, at near-zero marginal cost per use.

Why Did You Choose This Problem?



Why it's important

Communication is a basic human right. Closing this gap improves independence, safety, and dignity for millions in daily life.



Who benefits

Deaf & hard-of-hearing individuals, their families, educators, healthcare staff, and businesses serving diverse customers.



Why an AI Agent fits

Computer vision (gesture/pose recognition) + NLP + speech synthesis can run continuously, in real time, without a human interpreter present.



What makes it different

Combines live video input, gesture-to-text translation, and instant text-to-speech in one lightweight, mobile-first pipeline.



Potential impact

Scales to classrooms, hospitals, and public service desks — turning one-time interpreter costs into a reusable, low-cost AI service.



Real-Time Market Cost Breakdown

Estimated cost to build & run an MVP AI Sign Language Translator

Cost Item	Description	Estimated Cost
Dataset & model training	Sign-gesture dataset collection/licensing + training a pose-estimation & classification model	\$600 – \$3,000
Core app development	Camera capture, gesture recognition pipeline, UI/UX (2-person student team, part-time)	\$8,000 – \$25,000
Speech synthesis (TTS) API	Cloud text-to-speech for converting translated text into natural speech	\$4 – \$16 per 1M characters
Cloud compute / hosting	GPU inference + backend server for real-time video processing	\$50 – \$300 / month
Testing & maintenance	QA across devices, bug fixes, accuracy tuning post-launch	\$500 – \$2,000 / month

MVP BUILD (ONE-TIME)

\$9K – \$28K

student-team prototype

MONTHLY RUNNING COST

\$550 – \$2.3K

compute + TTS + upkeep

COST PER TRANSLATION

< \$0.01

far below a human interpreter

Sources: industry AI/translation-app development cost benchmarks (2025–2026); cloud TTS list pricing.